🔑 **Answer Key — Teacher Copy · Lesson 6-1**

MSTAR PRACTICE WORKSHEET

Lesson 6-1: Division Expressions with Fractions and Whole Numbers

Grade 6 Mathematics · Standard 6.NOS.1

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: A

What does 3 ÷ 1/4 mean?

- **A** How many 1/4-size pieces fit into 3
- **B** 1/4 of 3
- **C** 3 groups of 1/4
- **D** 3 minus 1/4

3 ÷ 1/4 asks how many 1/4-size pieces fit into 3. To find that, put a 1 under the 3 and use Keep, Change, Flip: 3/1 × 4/1 = 12.

Question 2 SELECTED RESPONSE — CORRECT: B

Which expression means 'how many 1/3-size pieces are in 2'?

- **A** 1/3 ÷ 2
- **B** 2 ÷ 1/3
- **C** 2 × 1/3
- **D** 2 + 1/3

Finding how many 1/3-size pieces fit into 2 is written 2 ÷ 1/3. Put a 1 under the 2, then Keep, Change, Flip: 2/1 × 3/1 = 6.

Question 3 SELECTED RESPONSE — CORRECT: D

A rope is 4 feet long. How many 1/2-foot pieces can be cut from it?

A 1/2 piece

B 2 pieces

C 4 pieces

D 8 pieces

> Write 4 ÷ 1/2. Put a 1 under the 4, then Keep, Change, Flip: 4/1 × 2/1 = 8 pieces.

Question 4 SELECTED RESPONSE — CORRECT: A

A half-pan of brownies is shared equally among 4 people. How much of the whole pan does each get?

A 1/8

B 1/2

C 2

D 4/2

> This is 1/2 ÷ 4. The whole number is second this time, so put a 1 under the 4, then Keep, Change, Flip: 1/2 × 1/4 = 1/8 of the whole pan.

Question 5 SELECTED RESPONSE — CORRECT: C

Agent Cole has 3/4 of a pound of fingerprint powder and splits it equally into 2 kits. How much powder goes in each kit?

A 1 1/2 pounds

B 1/8 pound

C 3/8 pound

D 3/4 pound

> Write 3/4 ÷ 2 as 3/4 ÷ 2/1. Keep, Change, Flip: 3/4 × 1/2 = 3/8 pound in each kit. The top number does not have to be a 1 — the steps are exactly the same.

Question 6 SELECTED RESPONSE — CORRECT: A

What is 2/3 ÷ 4?

A 1/6

B 2/7

C 3/8

D 8/3

Write the 4 as 4/1. Keep, Change, Flip: 2/3 × 1/4 = 2/12 = 1/6. Dividing by a whole number bigger than 1 makes the piece smaller, and 1/6 is smaller than 2/3.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: D

What is the value of 6 ÷ 1/5?

A 1/30

B 6/5

C 11

D 30

Put a 1 under the whole number so it becomes 6/1. Keep 6/1, change ÷ to ×, and flip 1/5 to its reciprocal 5/1. Multiply across: 6/1 × 5/1 = 30/1 = 30.

- **A:** That is the reciprocal of the answer. Keep the 6 on top: 6/1 times 5/1.
- **B:** That multiplies 6 by 1/5. Division flips the divisor first: 6/1 times 5/1.
- **C:** That adds 6 and 5. Keep, change, flip, and then multiply.

Part B — correct answer: B

Which reasoning shows why your answer to Part A is correct?

A Keep 1/5, change ÷ to ×, and flip 6/1 to 1/6: 1/5 × 1/6 = 1/30.

B Write 6 as 6/1. Keep 6/1, change ÷ to ×, flip 1/5 to 5/1, then multiply across: 6/1 × 5/1 = 30.

C Write 6 as 6/1 and multiply straight across without flipping: 6/1 × 1/5 = 6/5.

D Add the whole number and the denominator of the fraction: 6 + 5 = 11.

Keep, change, flip works only after the whole number is written as a fraction over 1, and only the second fraction gets flipped. That gives 6/1 × 5/1 = 30.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, C, E

Select ALL statements that are true about the expression 4 ÷ 1/3.

- **A** The whole number 4 can be written as the fraction 4/1.
- **B** Keep, change, flip rewrites the expression as 4/1 × 3/1.
- **C** The reciprocal of 1/3 is 3/1.
- **D** The quotient is 4/3.
- **E** The quotient is 12.
- **F** The quotient must be less than 4 because the expression uses division.

Write 4 as 4/1, keep it, change ÷ to ×, and flip 1/3 to 3/1. Then 4/1 × 3/1 = 12/1 = 12. The quotient 4/3 comes from multiplying by 1/3 instead of dividing, and dividing by a number smaller than 1 makes the quotient larger, not smaller.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

Dividing 6 by 1/2 gives 12, but dividing 6 by 2 gives 3. How can dividing by a smaller number (1/2) give a bigger answer than dividing by a larger number (2)? Explain using a real-world example.

Model answer: The two expressions ask different questions, and the flip shows why. 6 ÷ 2 is 6/1 × 1/2 = 3, because flipping 2/1 gives a factor smaller than 1. 6 ÷ 1/2 is 6/1 × 2/1 = 12, because flipping 1/2 gives a factor larger than 1. Dividing by a number less than 1 always makes the quotient grow.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.